

# Introduction to Map Projection

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with AI-assisted contribution from ChatGPT and  
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# Learning outcomes

After this lecture, students should be able to explain and choose projections for engineering work.

- Explain why every flat map must distort the globe.
- Distinguish conformal, equal-area, equidistant, and azimuthal properties.
- Relate projection choice to ship navigation, star maps, land mapping, military grids, and planetary mapping.
- Recognize why datum, coordinate system, and projection are a single workflow.

## Professor's message

A projection is not only a drawing method. It is an engineering decision about what error you can accept and what property you must preserve.

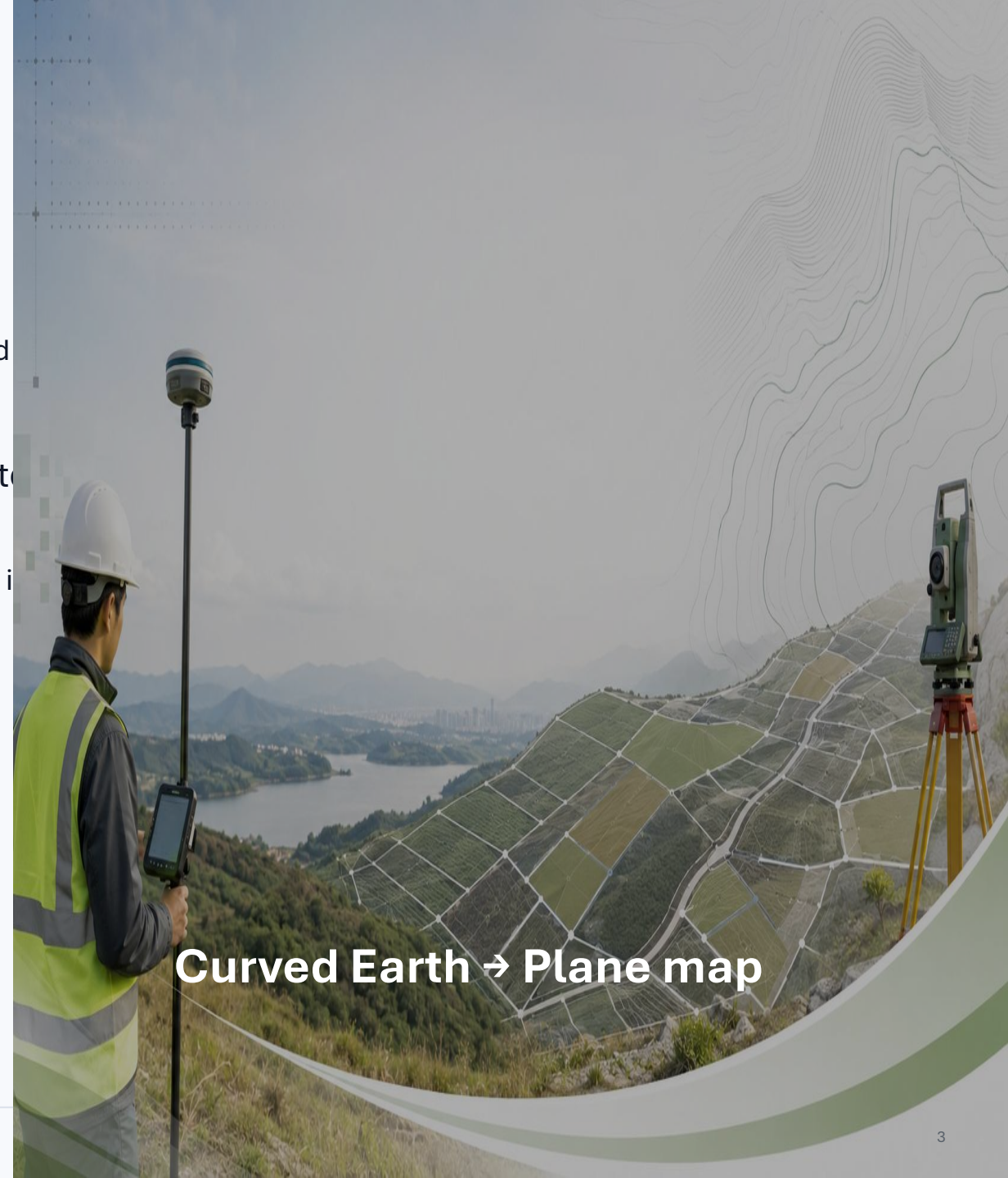
## Core question

For this map task, which quantity must be most reliable: shape, area, distance, direction, or local scale?

# Why map projection exists

Surveyors measure on a curved Earth but design, analyze, and construct on a plane.

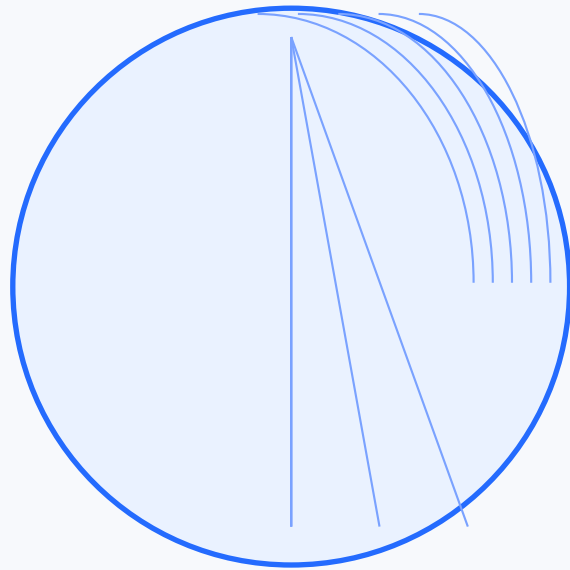
- A globe preserves geometry globally but is impractical for engineering drawings.
- A plane map supports coordinates, grids, CAD/GIS, route planning, and computation.
- Projection transforms latitude–longitude into x–y coordinates.
- The transformation always introduces distortion; the skill is controlling it.



**Curved Earth → Plane map**

# The basic transformation

Map projection is a mathematical function from geodetic coordinates to plane coordinates.



Geodetic  
longitude  $\lambda$   
latitude  $\phi$



**Projection  
formula**



Plane coordinates  
Easting  $x$   
Northing  $y$

## In formula form

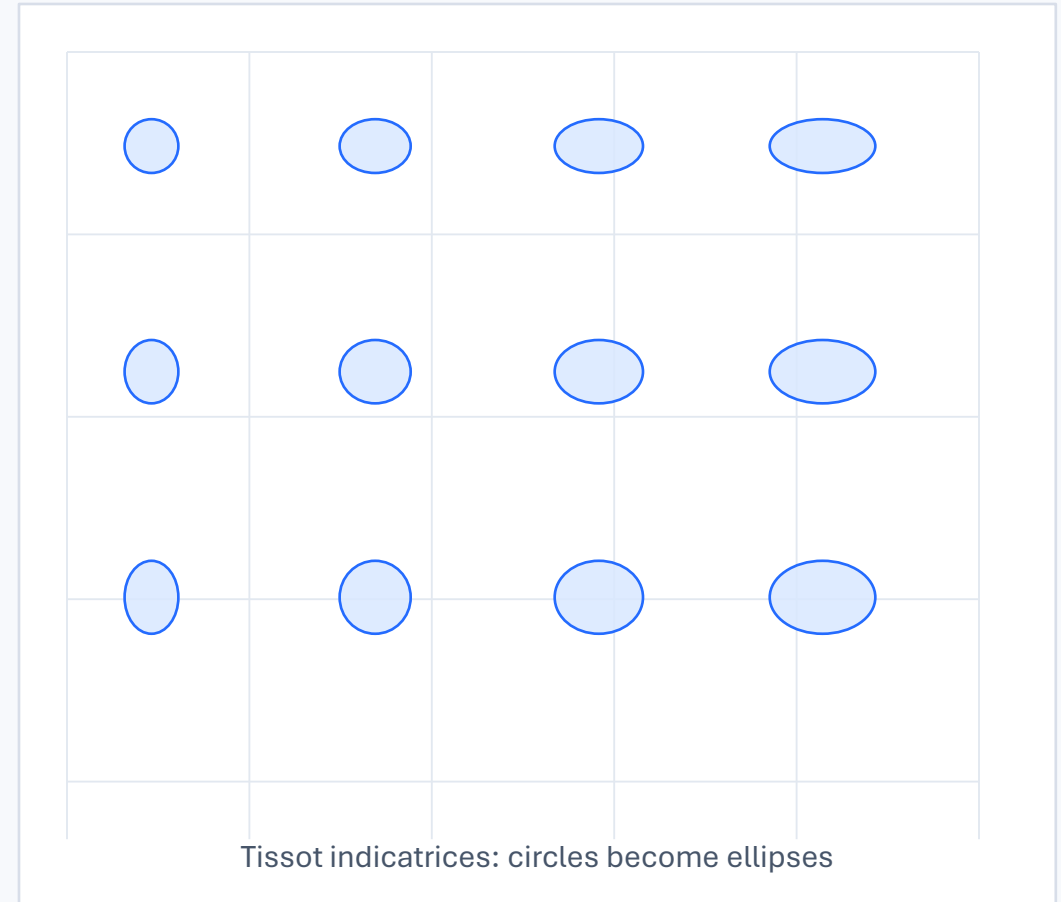
$$\begin{aligned}x &= f(\lambda, \phi) \\y &= g(\lambda, \phi)\end{aligned}$$

Projection is a coordinate transformation, not merely a picture.

# The unavoidable price: distortion

No flat map can preserve all distances, areas, directions, and shapes everywhere.

- Local scale varies from place to place and often by direction.
- Distortion can be visualized by Tissot indicatrices: small circles on Earth become ellipses on the map.
- A good projection puts the least distortion where the work is important.



# History of map projection

From star observation and exploration to modern geodesy

# Ancient foundations

Early cartography linked astronomy, navigation, and mathematics.

## Greek and Alexandrian tradition

Latitude, longitude, spherical Earth concepts, and gridded maps established the mathematical language of projections.

## Ptolemy's Geography

Ptolemy compiled coordinates and described map construction methods; conic-like ideas influenced later regional mapping.

## Instrumental legacy

Astrolabes, globes, and celestial charts connected sky positions with Earth positions.

- Projection began as a practical solution: draw the known world on a sheet.
- The same spherical geometry used for stars also supports Earth mapping.
- Survey engineering inherited both the astronomical and geodetic traditions.

# Renaissance and ocean navigation

The age of exploration created demand for maps that supported compass-course sailing.

- In 1569, Gerardus Mercator introduced a cylindrical conformal world map.
- The key value: constant-bearing routes appear as straight lines.
- The cost: area inflation toward high latitudes.
- This is an early example of choosing a projection for one dominant task.

## Engineering lesson

A “wrong-looking” map may be the correct tool when its preserved property matches the operation.



# 18th–19th century: projection science

Mathematicians formalized conformal, equal-area, and geodetic mapping methods.

## Lambert

Conformal conic and azimuthal equal-area ideas supported regional and thematic mapping.

## Gauss / Krüger

Transverse Mercator mathematics became central for large-scale national and engineering coordinate systems.

## Geodesy

Ellipsoids, datums, triangulation networks, and precise surveys demanded rigorous coordinate transformation.

- The map became a computational surface, not just an artistic product.
- Projection, datum, and survey control developed together.
- Large-scale mapping prefers local accuracy over global appearance.

# 20th–21st century: GIS, GNSS, and web maps

Modern projection choice is embedded in software, databases, and real-time positioning.

- Topographic maps: national grids and UTM zones.
- GIS: data layers must share CRS or transform correctly.
- GNSS: ellipsoidal positions need projection and height treatment before engineering use.
- Web maps: fast global visualization may not be suitable for precise measurement.

## Today's problem

Students often click “reproject” in GIS without checking distortion, datum, units, and area of use.

## Professional habit

Before measurement or design: check CRS name, EPSG code, datum, projection method, zone, unit, and scale factor.

# Projection families

Most projections can be understood by the developable surface or geometric concept.

## Cylindrical

Best conceptual model for equatorial worlds and Mercator-style navigation. Meridians and parallels often form a rectangular grid.

## Conic

Useful for mid-latitude regions wider east–west. Standard parallels control scale distortion.

## Azimuthal

Plane touches or cuts the globe. Useful for polar regions, radio/radar range, and sky charts.

## Pseudo / compromise

Designed for visual balance, atlases, and thematic maps; they do not preserve one exact property everywhere.

# Projection properties: what can be preserved?

Projection choice begins by naming the property that matters most.

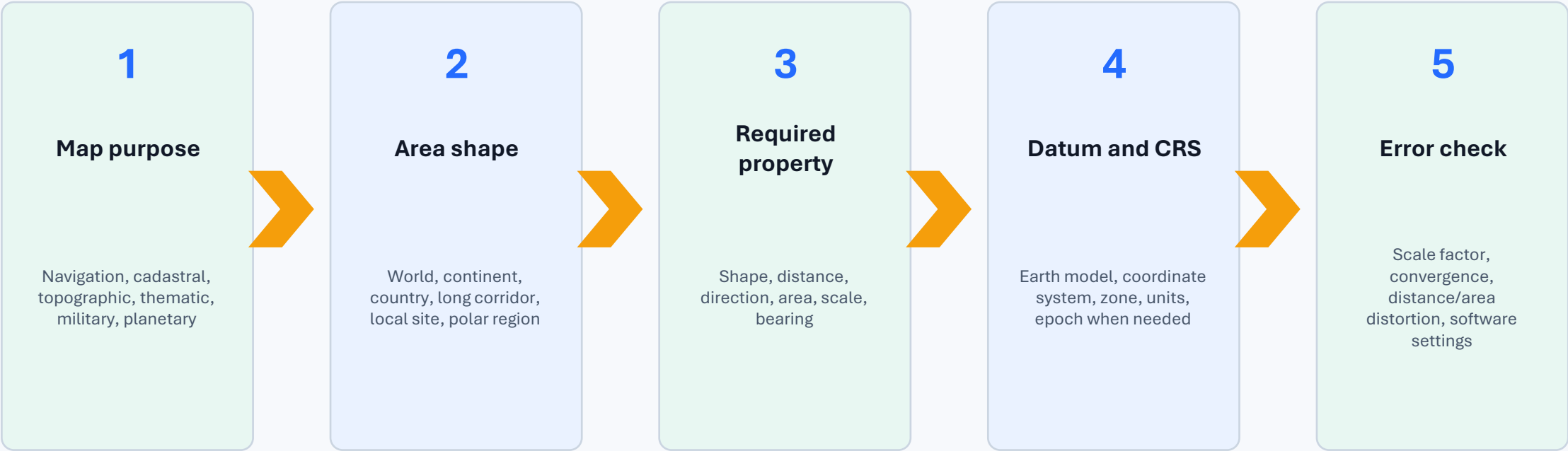
| Property    | Preserves                           | Common use                             |
|-------------|-------------------------------------|----------------------------------------|
| Conformal   | Local shape / angle                 | Cadastral, engineering, navigation     |
| Equal-area  | Area                                | Thematic, resource, environmental maps |
| Equidistant | Distance from selected lines/points | Route/range maps                       |
| Azimuthal   | Direction from center               | Radar, polar, sky maps                 |

## Important

No projection can preserve all four globally. A conformal map is not equal-area; an equal-area map changes local shape.

# How to choose a projection: a professional workflow

Projection selection is an engineering design step.



## Field rule

Never evaluate projection only from the map's visual appearance. Evaluate it from the measurement task.

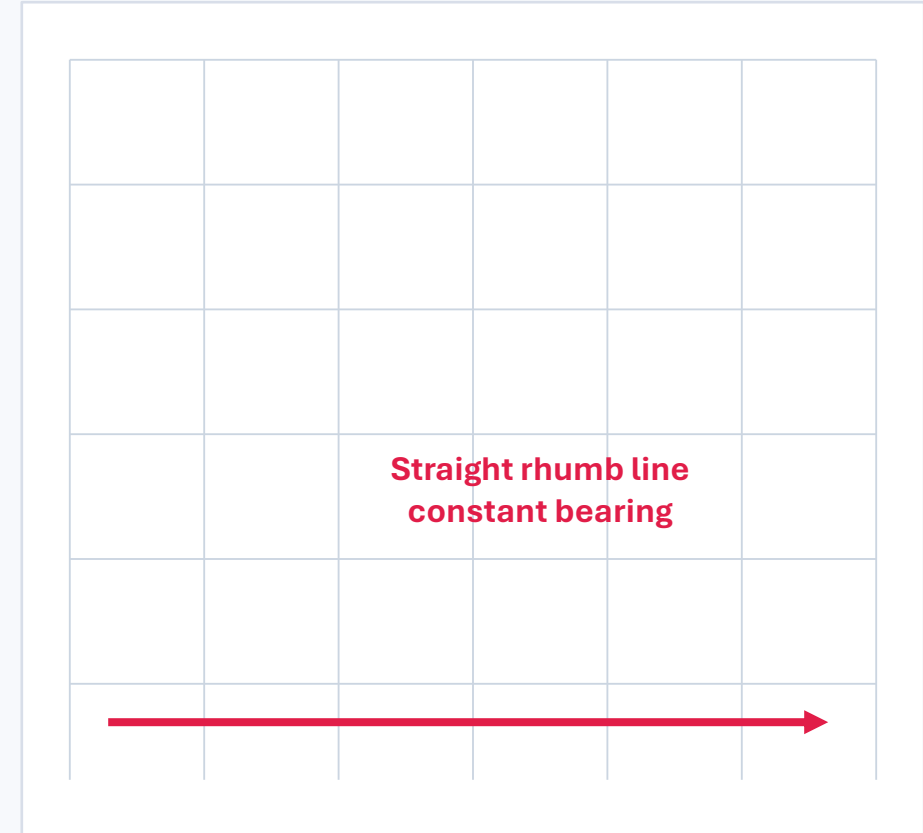
# Application: ship navigation

Mercator, rhumb lines, great circles, and marine charts

# Why Mercator became important for ships

Marine navigation values bearings and course plotting.

- Most nautical charts use Mercator projection for practical course plotting.
- On a Mercator chart, a rhumb line is straight: constant compass bearing.
- The navigator can draw a line and measure course angle against meridians.
- At high latitudes, scale and area increase strongly; polar navigation uses alternatives.



# Rhumb line vs great circle

The shortest route on the globe is not usually a straight line on a Mercator chart.

## Rhumb line

Constant bearing. Easy to steer historically and easy to plot on Mercator charts, but not always shortest.

## Great circle

Shortest path on a spherical/ellipsoidal Earth. On Mercator it appears curved except for special cases.

## Modern navigation

GNSS, ECDIS, and route optimization allow great-circle planning while still using projected charts for display and local control.



# Navigation lesson for survey engineers

A projection can encode an operational method.

- A sea chart is not only a base map; it is a working navigation instrument.
- Course, bearing, scale, and chart datum must be checked together.
- Projection choice may change with latitude, scale, and route length.
- Modern systems hide many transformations; professionals must still understand them.

## Discussion question

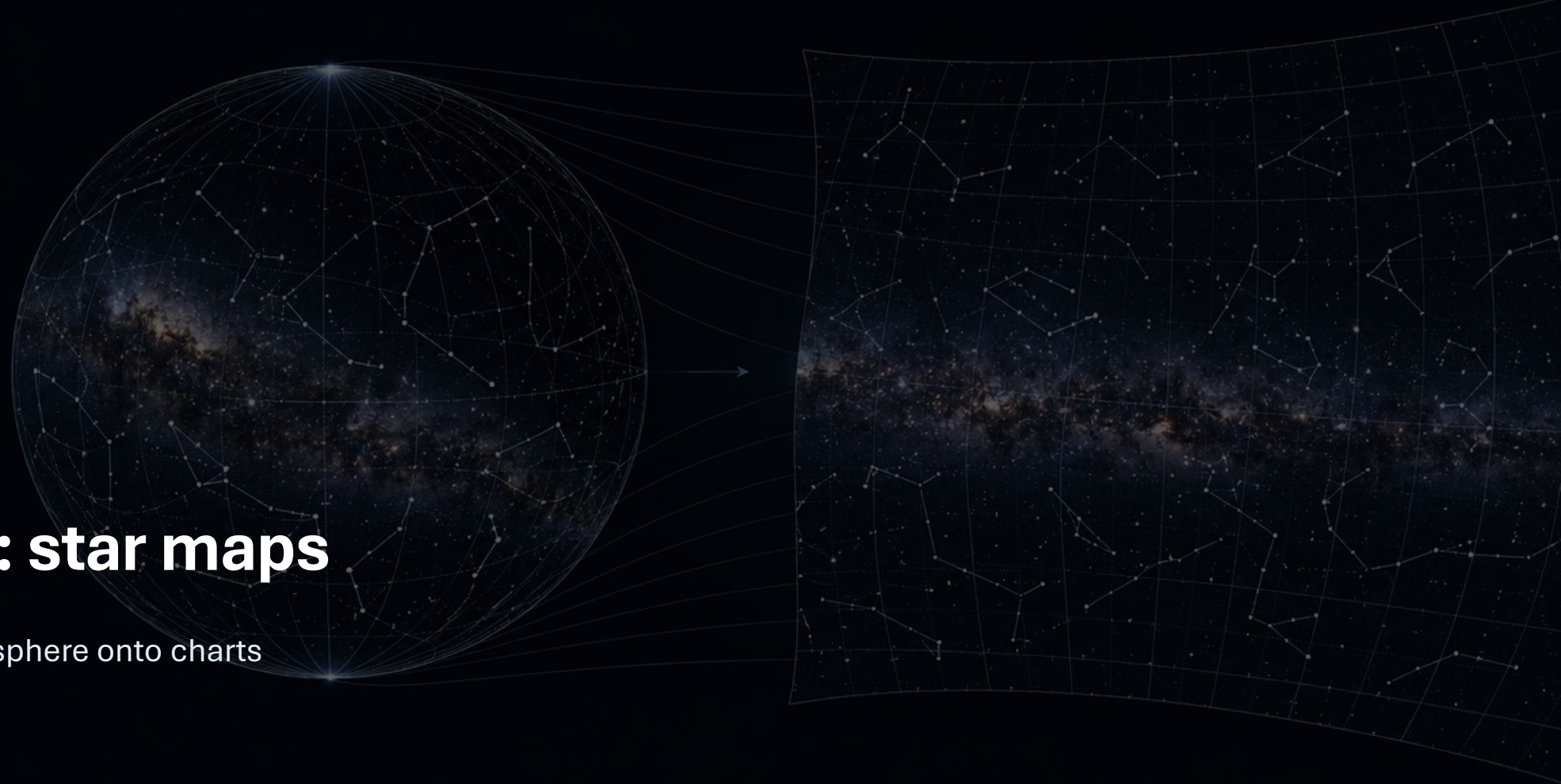
For an autonomous vessel, which projection would you use for: global route planning, coastal approach, and harbor maneuvering?

## Survey analogy

For a road corridor, a pipeline, or a GNSS baseline, the “best” projection may be a custom local projection rather than a standard national map.

# Application: star maps

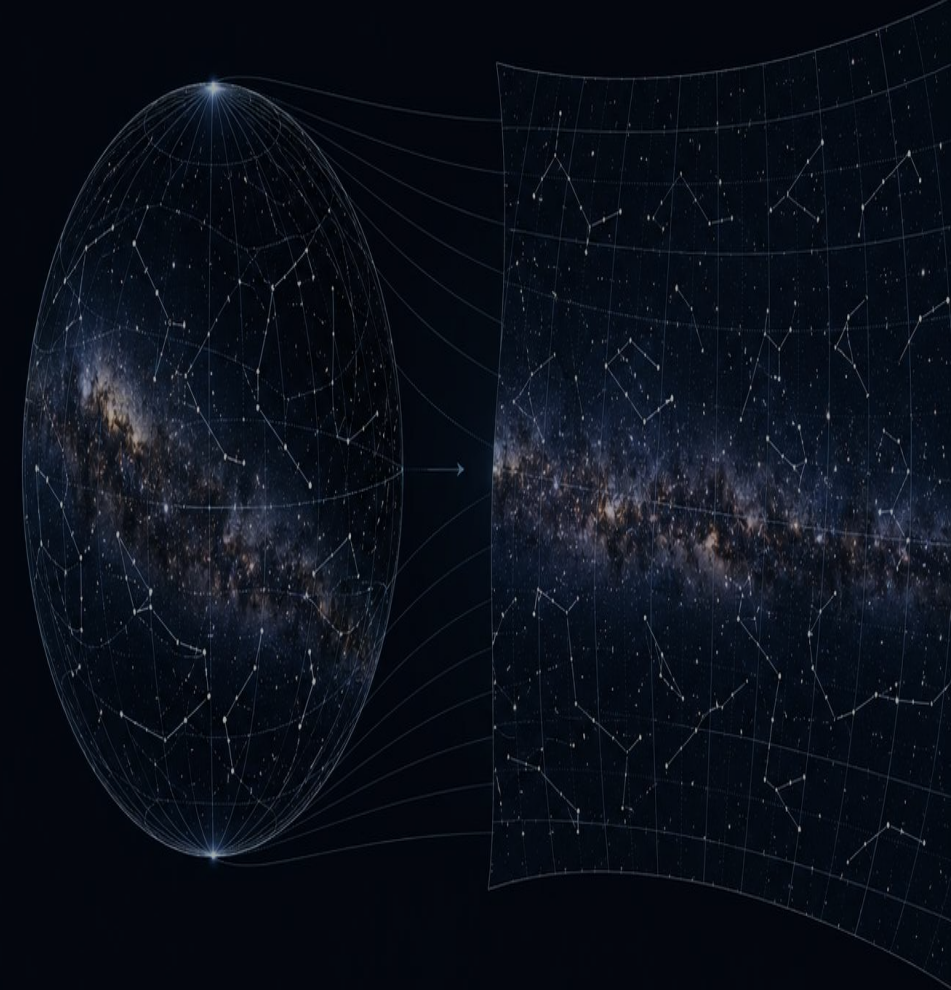
Projecting the celestial sphere onto charts



# Star maps: the sky is also a sphere

Astronomy uses coordinates on the celestial sphere, then projects them onto a chart.

- Stars are located by right ascension and declination, analogous to longitude and latitude.
- Full-sky maps need projection because a dome cannot be printed flat.
- The projection changes how constellations look near chart edges or poles.
- For observing, the best chart depends on sky region and field of view.



# Common star-chart projection ideas

The same geometry appears in Earth maps and sky maps.

## Stereographic

Conformal; useful for planispheres and polar sky charts because local angles are preserved.

## Gnomonic

Great circles plot as straight lines; useful conceptually for navigation and some astronomical charts.

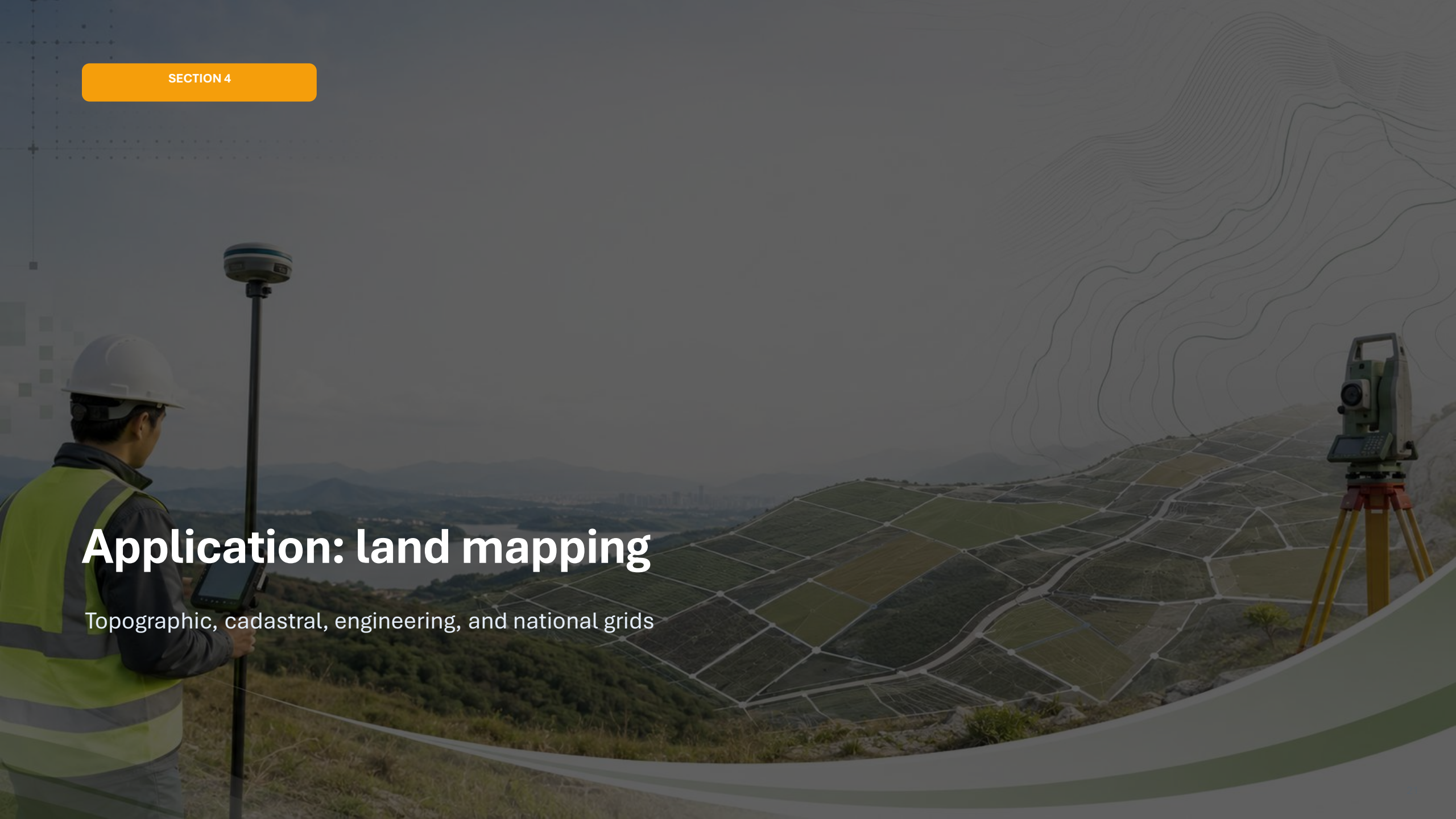
## Azimuthal equidistant

Distances from the center are preserved; useful for circular all-sky views centered on the observer.

- A celestial map is a projection of direction vectors, not a physical land surface.
- Distortion affects constellation shape, angular distance, and visual interpretation.
- Projection literacy helps students read sky maps and satellite sensor geometry.

# Application: land mapping

Topographic, cadastral, engineering, and national grids



# Land mapping: accuracy is local and legal

Survey engineering uses projections to turn observations into authoritative coordinates.

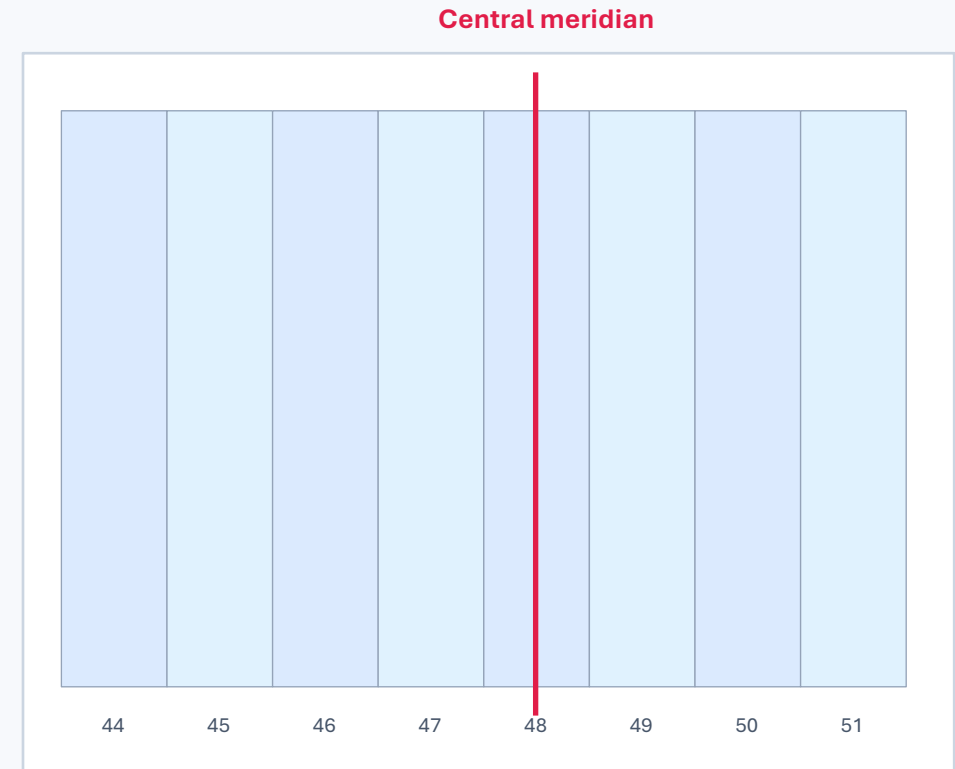
- Topographic mapping needs consistent national coverage.
- Cadastral mapping needs local shape, angle, and area traceability.
- Engineering design needs low distortion over project corridors and sites.
- Projection parameters become part of the legal and technical record.



# UTM: a workhorse for land mapping

Universal Transverse Mercator divides Earth into narrow zones to control distortion.

- Transverse Mercator cylinder is oriented around a central meridian.
- Narrow 6° zones keep scale distortion manageable for mapping.
- Coordinates are in meters, supporting GIS and engineering workflows.
- Zone boundaries and datum differences must be handled carefully.



6° zones with meter coordinates

# Local projections for engineering

For high-precision work, surveyors often reduce distortion with a local design.

## Problem

A standard projection may introduce grid-ground distance differences that matter for construction.

## Solution

Use local datum/projection parameters, combined scale factor, or low-distortion projection design.

## Check

Compare geodetic distance, grid distance, elevation factor, and scale factor before setting out.

- Large-scale engineering cares about centimeters, not only map appearance.
- Projection scale factor and ground-to-grid correction must be documented.
- Coordinate epoch may matter when using modern GNSS and national CORS networks.

# Application: military mapping

Grid reference, terrain, logistics, and interoperability

# Military mapping: speed, grid, and shared re

The map must support rapid communication under pressure.

- Grid coordinates allow concise position reporting and coordination.
- Terrain maps support visibility, movement, logistics, and risk analysis.
- Projection choice influences distance, direction, and grid convergence.
- Interoperability requires common datum, grid, symbology, and metadata.



# Military grid concepts

A projection turns Earth position into operational grid references.

## Grid reference

A short coordinate language for reporting locations, routes, and areas.

## Grid convergence

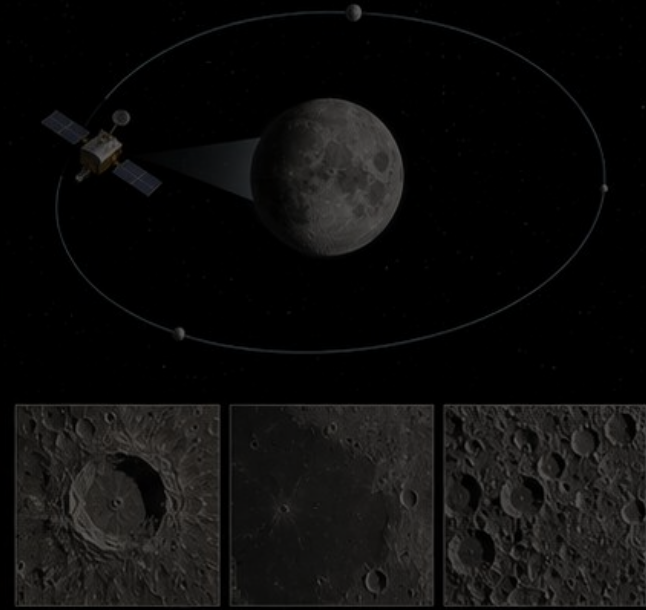
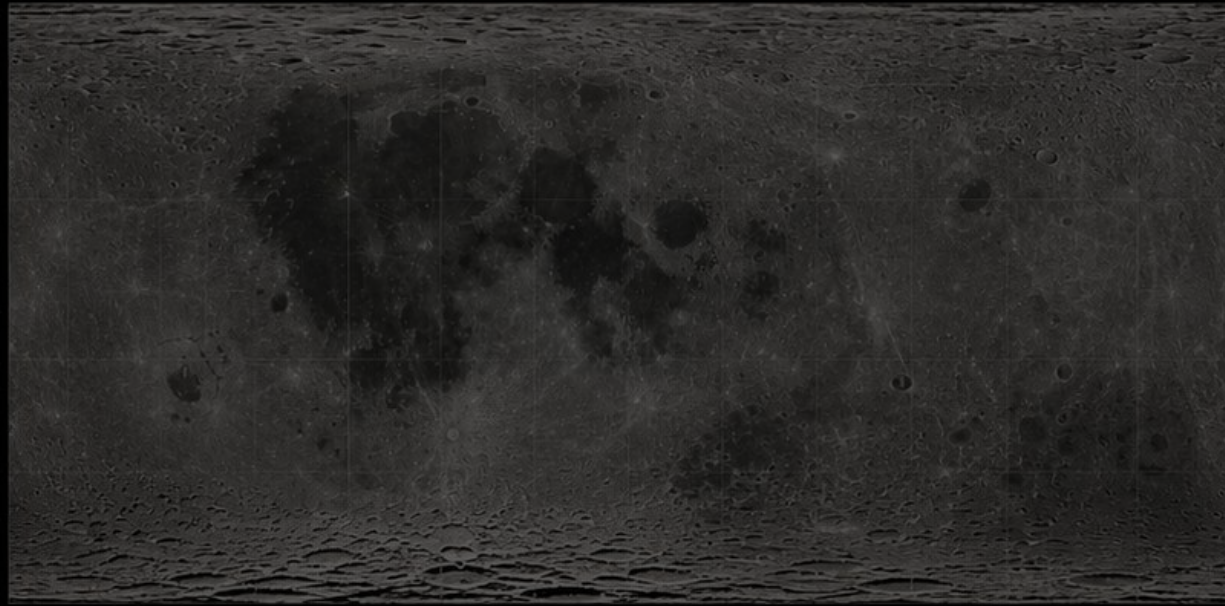
Angle between grid north and true north; important for bearings and artillery/sensor orientation.

## Scale and distance

Ground, geodetic, and grid distances are not identical; high-precision targeting and logistics require correction.

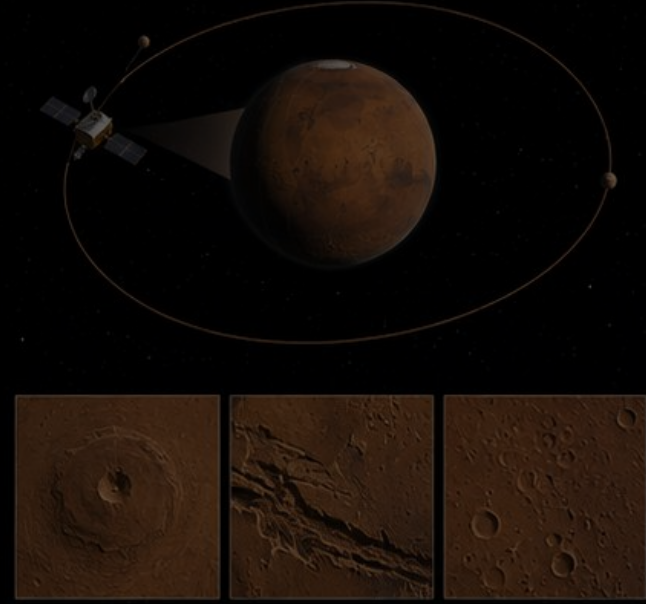
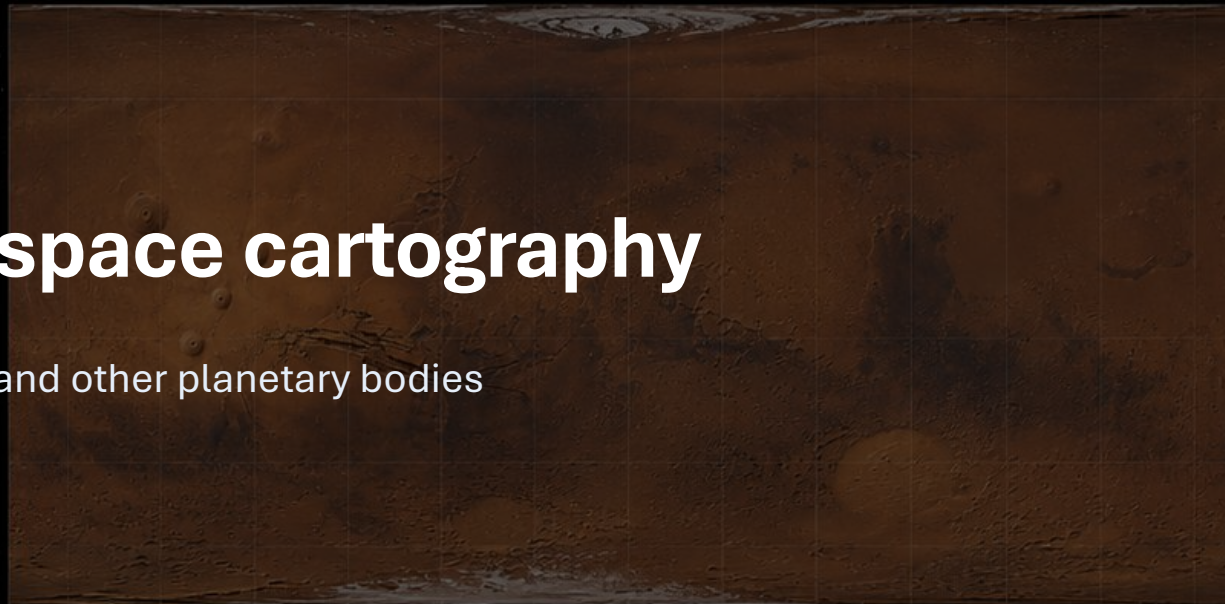
- Military use emphasizes a shared, unambiguous reference system.
- The same map may serve both human reading and machine navigation.
- Datum mistakes can be more dangerous than cartographic distortion.

SECTION 6



# Application: space cartography

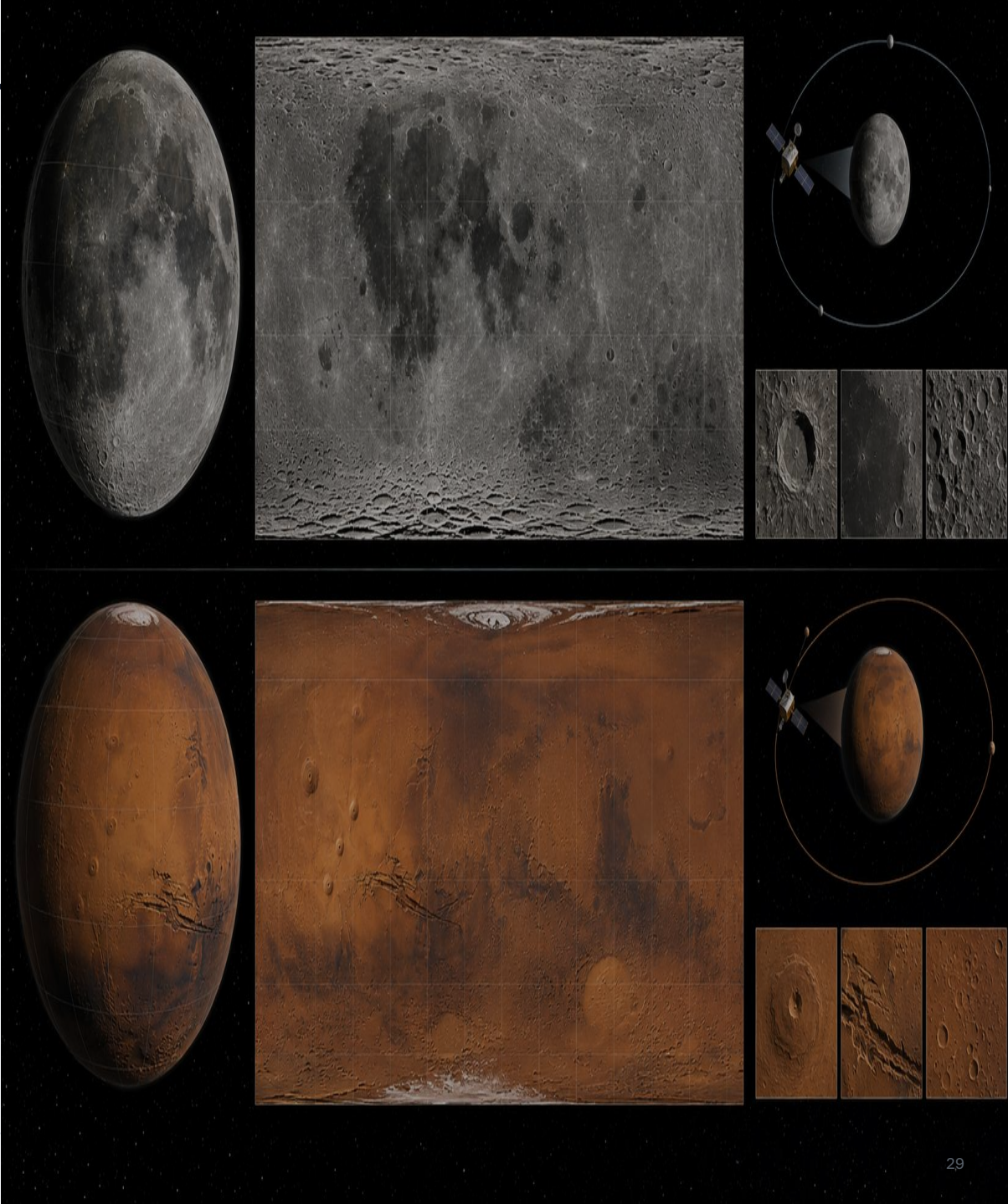
Mapping the Moon, Mars, and other planetary bodies



# Moon and Mars: map projection beyond Earth

Planetary cartography uses the same projection logic with different bodies.

- A “datum” must define body shape, rotation axis, prime meridian, and coordinate direction.
- Moon and Mars maps support landing-site selection, traverse planning, geology, and hazard assessment.
- Different surface models and reference frames affect coordinates and distances.
- Projection must match the region: global mosaic, polar landing area, crater map, or rover traverse.



# Key takeaways and sources

Projection choice is a professional engineering decision.

- A projection transforms curved coordinates to plane coordinates; distortion is unavoidable.
- Choose projection by purpose: navigation, sky mapping, land survey, military operations, or planetary exploration.
- Always document datum, projection method, parameters, unit, zone, and area of use.
- For survey engineering, validate local scale and grid-ground relationships before measurement or design.

## Selected sources

USGS: Map Projections — A Working Manual; USGS: Map projections introduction.

NOAA: Nautical cartography and Mercator chart use.

NASA SVS: Deep Star Maps and Gaia-based star-map data.

IAU: coordinate systems for Solar System bodies.

NASA/USGS Astrogeology: planetary mapping and Astropedia catalog.

## Student exercise

Select a project area in Thailand. Compare UTM Zone 47N/48N and one local low-distortion projection. Report scale factor and distance difference for a 5 km line.